Bill of Materials - HMI Subsystem

Part Name/Description	Unit Quantity	Unit Prototype Cost	Total Prototype Cost	Unit Production Cost	Total Production Cost	Manufacturer	Manufacturer Part #	Vendor Link	Datasheet Link	Supplier	Supplier Part #	# Ordered	Date Ordered	# Received	Surplus	Schematic Reference Designators
ESP32-S3-WROOM-1 Module, SMD, 4MB Flash	1	$0.00	$0.00	$0.00	$0.00	Espressif	ESP32-S3-WROOM-1-N	https://www.digikey.com/en/products/detail/espressif-systems/ES		Peralta Labs	1965-ESP32-S3-WROOM-1-N4CT-N	0		0	-1	E1
Buck Switching Regulator, 3.3V Fixed, 3A, TO-263-5	1	$6.97	$6.97	$6.97	$6.97	Texas Instruments	LM2596S-3.3/NOPB	https://www.digikey.com/en/products/detail/texas-instruments/LA		Peralta Labs	LM2596S-3.3/NOPB-ND	0		0	-1	U1
USB Mini B Receptacle, 5-Pin, SMT Right Angle	1	$0.00	$0.00	$0.00	$0.00	Hirose	UX60SC-MB-5S8	https://www.digikey.com/en/products/detail/hirose-electric-co-ltd		Peralta Labs	H11645-ND	0		0	-1	J1
Connector Header, Through Hole, 8 Position, 2.54mm	5	$0.00	$0.00	$0.00	$0.00	Wurth Elektronik	61300811121	https://www.digikey.com/en/products/detail/w%C3%BCrth-elektro		Peralta Labs	609-5123-ND	0		0	-5	J3, J4, J5, J6, J7
DC Barrel Jack with Switch, 2.1mm ID, PCB Mount	1	$0.64	$0.64	$0.64	$0.64	Same Sky (CUI Device	PJ-002BH	https://www.digikey.com/en/products/detail/cui-devices/PJ-002BH		DigiKey	CP-002BH-ND	0		0	-1	J2
0.96" I2C OLED Display Module, 128x64, SSD1306	1	$0.00	$0.00	$0.00	$0.00	Songhe	B085WCRS7C	https://www.amazon.com/dp/B085WCRS7C		Class	-	0		0	-1	LCD1
Schottky Rectifier Diode, 5A, 40V, DO-214AB SMD (1N582	1	$0.42	$0.42	$0.42	$0.42	Vishay	SSC54-E3/57T	https://www.digikey.com/en/products/detail/vishay-general-semic		DigiKey	SSC54-E3/57TGICT-ND	0		0	-1	D1
ESD Protection TVS, 5.25V, USB 2.0 HS, SOT-23-6	2	$0.46	$0.92	$0.46	$0.92	STMicroelectronics	USBLC6-2SC6Y	https://www.digikey.com/en/products/detail/stmicroelectronics/U		DigiKey	497-11882-2-ND	0		0	-2	D6, D7
LED Blue, T-1 3/4 (5mm), 20mA	1	$0.00	$0.00	$0.00	$0.00	Wurth Elektronik	151051BS04000			Peralta Labs	732-5013-ND	0		0	-1	D2
LED Green, T-1 3/4 (5mm), 20mA	1	$0.00	$0.00	$0.00	$0.00	Wurth Elektronik	151051VS04000			Peralta Labs	732-5015-ND	0		0	-1	D3
LED White, T-1 3/4 (5mm), 20mA	1	$0.00	$0.00	$0.00	$0.00	Wurth Elektronik	151051WS04000			Peralta Labs	732-5016-ND	0		0	-1	D4
LED Red, T-1 3/4 (5mm), 20mA	1	$0.00	$0.00	$0.00	$0.00	Wurth Elektronik	151051RS04000			Peralta Labs	732-5011-ND	0		0	-1	D5
Fixed Inductor, 33uH, 3A, Shielded SMD	1	$1.01	$1.01	$1.01	$1.01	Bourns	SRR1260-330M	https://www.digikey.com/en/products/detail/bourns-inc/SRR1260		DigiKey	SRR1260-330MCT-ND	0		0	-1	L1
Fuse, 1A Fast-Blow, 5x20mm Glass Cartridge	1	$0.53	$0.53	$0.53	$0.53	Littelfuse	0217001.MXP	https://www.digikey.com/en/products/detail/littelfuse-inc/021700		DigiKey	F5929-ND	0		0	-1	F1
Fuse Holder, 5x20mm Cartridge, PCB Mount	1	$0.62	$0.62	$0.62	$0.62	Wurth Elektronik	696108003002	https://www.digikey.com/en/products/detail/w%C3%BCrth-elektro		DigiKey	732-11376-ND	0		0	-1	F1 holder
Electrolytic Capacitor, 100uF 16V, Radial Can SMD	1	$0.25	$0.25	$0.25	$0.25	Panasonic	ECA-1CM101	https://www.digikey.com/en/products/detail/panasonic-electronic-		DigiKey	P5138-ND	0		0	-1	C1
Electrolytic Capacitor, 220uF 16V, Radial Can SMD	1	$0.53	$0.53	$0.53	$0.53	Panasonic	EEE-1AA221XAP	https://www.digikey.com/en/products/result?keywords=EEE-1AA2		DigiKey	PCE4570CT-ND	0		0	-1	C2
Ceramic Capacitor, 0.1uF 50V, 0805 SMD (x3)	3	$0.00	$0.00	$0.00	$0.00	—	—			Peralta Labs	Lab Reels	0		0	-3	C3, C4, C5
Resistor, 10k, 1206 SMD (x7)	7	$0.00	$0.00	$0.00	$0.00	—	—			Peralta Labs	Lab Reels	0		0	-7	R1, R2, R3, R4, R5, R6, R7
Resistor, 10k, 1206 SMD (x3)	3	$0.00	$0.00	$0.00	$0.00	—	—			Peralta Labs	Lab Reels	0		0	-3	R10, R11, R16
Resistor, 27 Ohm, 1206 SMD (x2)	2	$0.00	$0.00	$0.00	$0.00	—	—			Peralta Labs	Lab Reels	0		0	-2	R8, R9
Resistor, 470 Ohm, 1206 SMD (x2)	2	$0.00	$0.00	$0.00	$0.00	—	—			Peralta Labs	Lab Reels	0		0	-2	R12, R13
Resistor, 100 Ohm, 1206 SMD (x4)	4	$0.00	$0.00	$0.00	$0.00	—	—			Peralta Labs	Lab Reels	0		0	-4	R14, R15, R17, R18
Tactile Push Button Switch, SPST, Through Hole (x10)	10	$0.18	$1.80	$0.18	$1.80	E-Switch	TL1105SPF160Q	https://www.digikey.com/en/products/detail/e-switch/TL1105SPF		DigiKey	EG2506-ND	0		0	-10	SW1, SW2, SW3, SW4, SW5, SW6, SW7, SW8, SW9, SW10
2-Pin Jumper, 2.54mm (x3)	3	$0.12	$0.36	$0.12	$0.36	Wurth Elektronik	61300211121	https://www.digikey.com/en/products/detail/wurth-elektronik/61		DigiKey	732-5887-1-ND	0		0	-3	JP1, JP2, JP3
Schottky Diode, 30V 200mA, SOD-123	1	$0.17	$0.17	$0.17	$0.17	onsemi	BAT54T1G	https://www.digikey.com/en/pro	https://www.onsemi.com/pdf/da	DigiKey	BAT54T1GOSDKR-ND					D8
Fuse, 500mA Fast-Blow, 5x20mm Glass Cartridge	1	$0.53	$0.53	$0.53	$0.53	Littelfuse	0217.500MXP	https://www.digikey.com/en/pro	https://www.littelfuse.com/~/me	DigiKey	F4744-ND					F2
Test Point, SMD Pad (x30)	30	$0.00	$0.00	$0.00	$0.00	—	—			PCB (no vendo	-	0		0	-30	TP1-TP30
TOTALS			**$14.75**		**$14.75**											